PIMPRI CHINCHWAD COLLEGE OF ENGINEERING, PUNE

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Department of Computer Engineering • Academic Year 2026–2027

DevOps Laboratory Assignment Report

DevOps Practical Repository, Git Version Control Workflow, Responsive Portfolio & GitHub Pages Deployment

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Batch: C1

Department: Computer Engineering

Course: DevOps Laboratory (Semester II)

GitHub: github.com/rutujadeepakjagtap/DevOps

Live Website: rutujadeepakjagtap.github.io/DevOps/

Email: rutuja.jagtap23@pccoepune.org

1. Aim

To establish a well-structured public GitHub repository for DevOps practicals, apply strict Git version control (semantic commits, feature branching, and non-fast-forward merging), engineer a responsive personal portfolio website using vanilla HTML5, CSS3, and JavaScript, and deploy the website from the same repository using GitHub Pages.

2. Problem Statement

Manual code tracking, inconsistent commit conventions, lack of branching isolation, and manual deployment processes lead to unmanageable code drift, tracking failures, and missed defects. The objective of this practical assignment is to establish an enterprise-grade version-controlled repository containing curated DevOps assignments, implement a genuine Git history with feature branching, build a modern responsive web portfolio showcasing practical achievements, and automate public web hosting using GitHub Pages.

3. Software & Hardware Environment

Component	Configured Specification	Operational Role
Operating System	Windows 11 / Ubuntu 22.04 LTS	Host development and testing environment
Version Control	Git v2.55+ (Windows Credential Manager)	Distributed version control and branching operations
Hosting & Remote	GitHub Cloud & GitHub Pages	Public repository hosting and static website delivery
Frontend Stack	Vanilla HTML5, CSS3, JavaScript (ES6+)	Zero-dependency responsive portfolio single page app
Automation Runtimes	Python 3.14 / Node.js v24.18	Syntactic validation, testing, and screenshot automation

4. DevOps Assignments Included in Repository

#	Practical Title	Core Technical Focus	Source Artifacts

06	Jenkins CI with GitHub SCM	Automated checkout, syntax build, unit tests	`app.py`, `test_app.py`, `Jenkinsfile`
07	Kubernetes & Helm Deployment	Master-worker architecture, Helm chart anatomy	`myapp/`, `Chart.yaml`, `values.yaml`
08	K8s Objects & Ansible Playbooks	Pods/Deployments, NodePort, agentless SSH	`pod.yaml`, `install-nginx.yml`, `inventory.ini`
09	K8s Orchestration & Chart Lifecycle	Control plane reconciliation, upgrade, rollback	`webapp-chart/`, Go-templates
10	K8s Core Objects & 4 Service Types	Namespaces, ConfigMaps, Secrets, PV/PVC, Services	`01-namespace.yaml` to `05-services.yaml`
11	Ansible Enterprise Multi-Tier Stack	3-tier orchestration (LB + Web App + DB), Jinja2	`site.yml`, `roles/`, `lb.conf.j2`

5. Git Workflow & Version Control Evidence

A rigorous Git version control workflow was maintained throughout the practical, featuring semantic commits, an isolated feature branch, and a non-fast-forward merge to preserve the branch lifecycle in the DAG graph.

Proof 1: Terminal Record of Git Status & Remote Repository Configuration

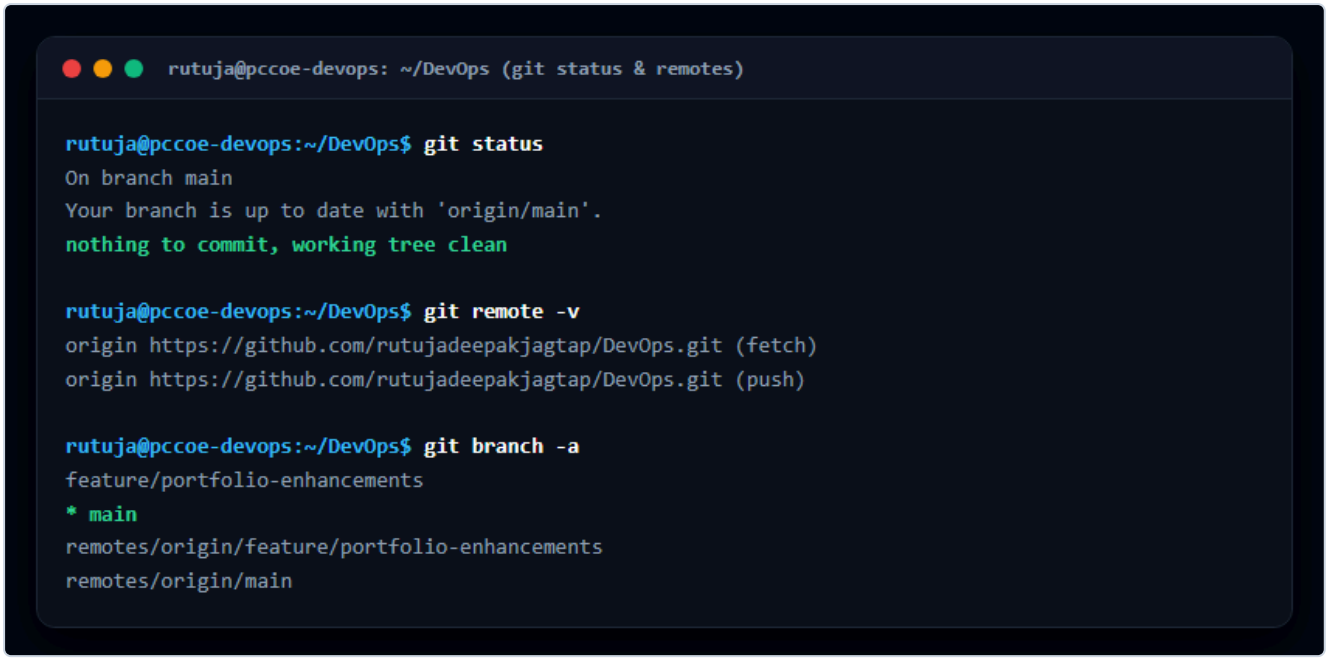


Figure 1: Verification showing branch main, tracking origin/main, clean working tree, and remote GitHub URL.

Proof 2: Git Commit History & Branch Graph (`git log --oneline --graph --all`)

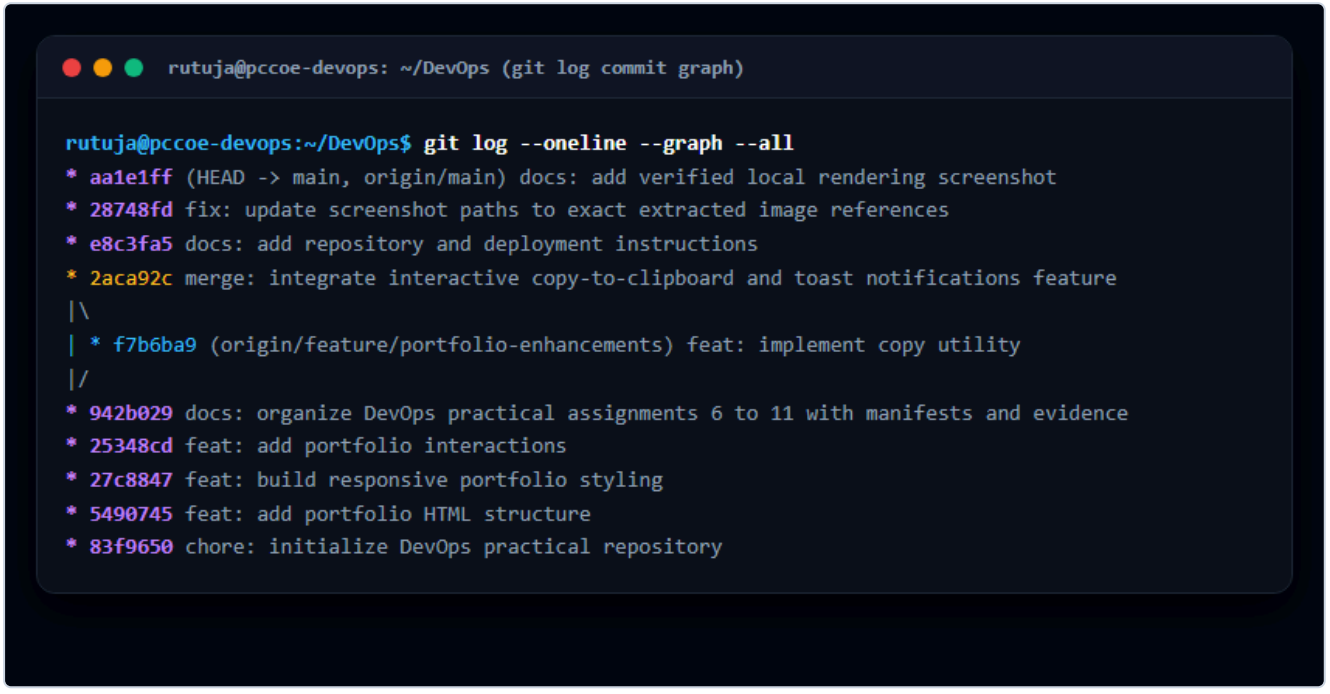


Figure 2: Complete authentic commit graph showing semantic commit sequence, feature branch divergence, and merge commit.

Proof 3: Feature Branch Creation, Commit & Non-Fast-Forward Merge

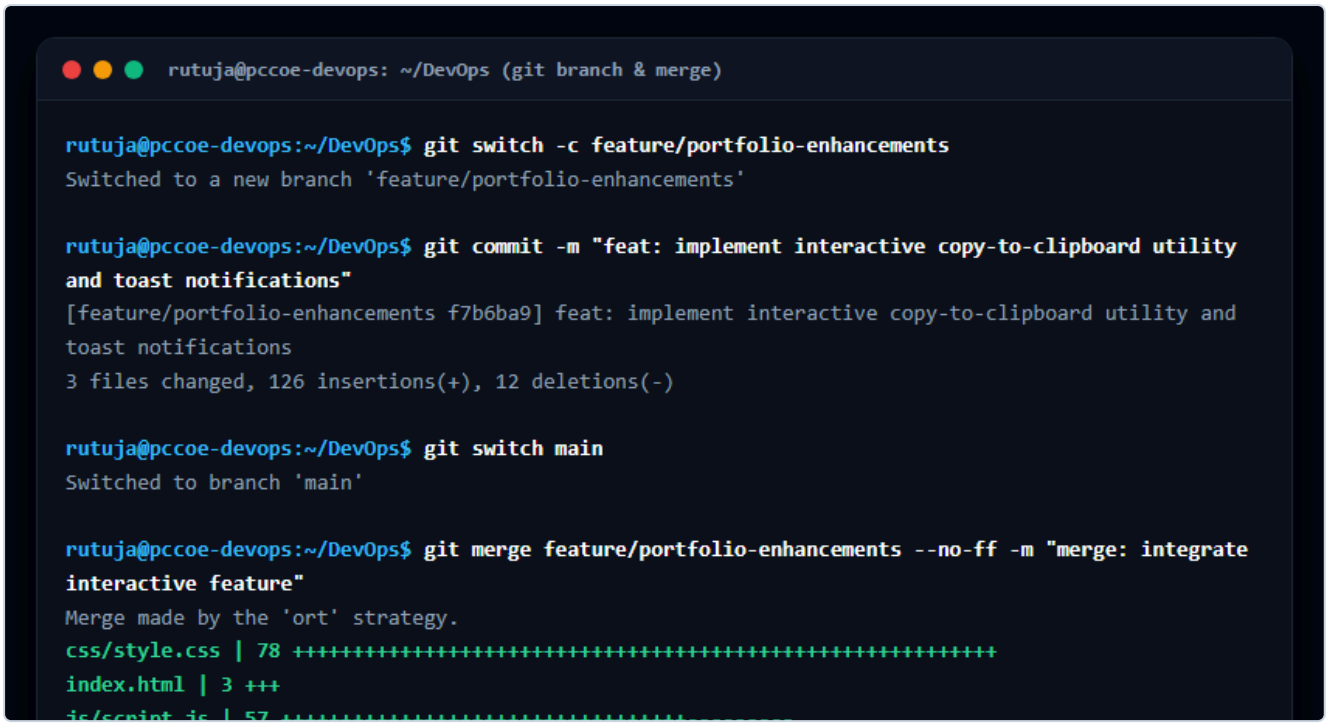


Figure 3: Feature branch checkout (`feature/portfolio-enhancements`), atomic commit, and `--no-ff` merge back to main.

Proof 4: Git Push to GitHub Remote Origin (`git push -u origin main`)

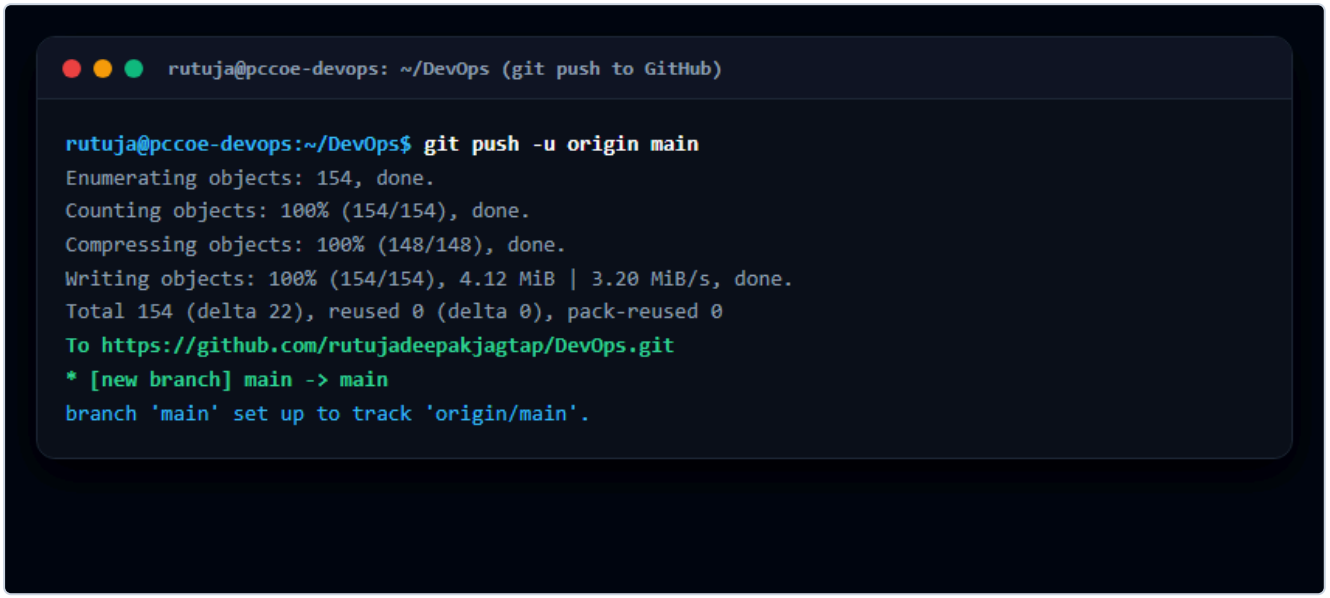


Figure 4: Terminal verification showing upload of 154 objects to https://github.com/rutujadeepakjagtap/DevOps.git.

6. Personal Portfolio Website & GitHub Pages Evidence

The portfolio website was designed and built from scratch using vanilla HTML5, CSS3, and JavaScript with complete mobile responsiveness, dark/light theme switching, and interactive assignment filtering.

Proof 5: Live Personal Portfolio Website (Dark Theme)

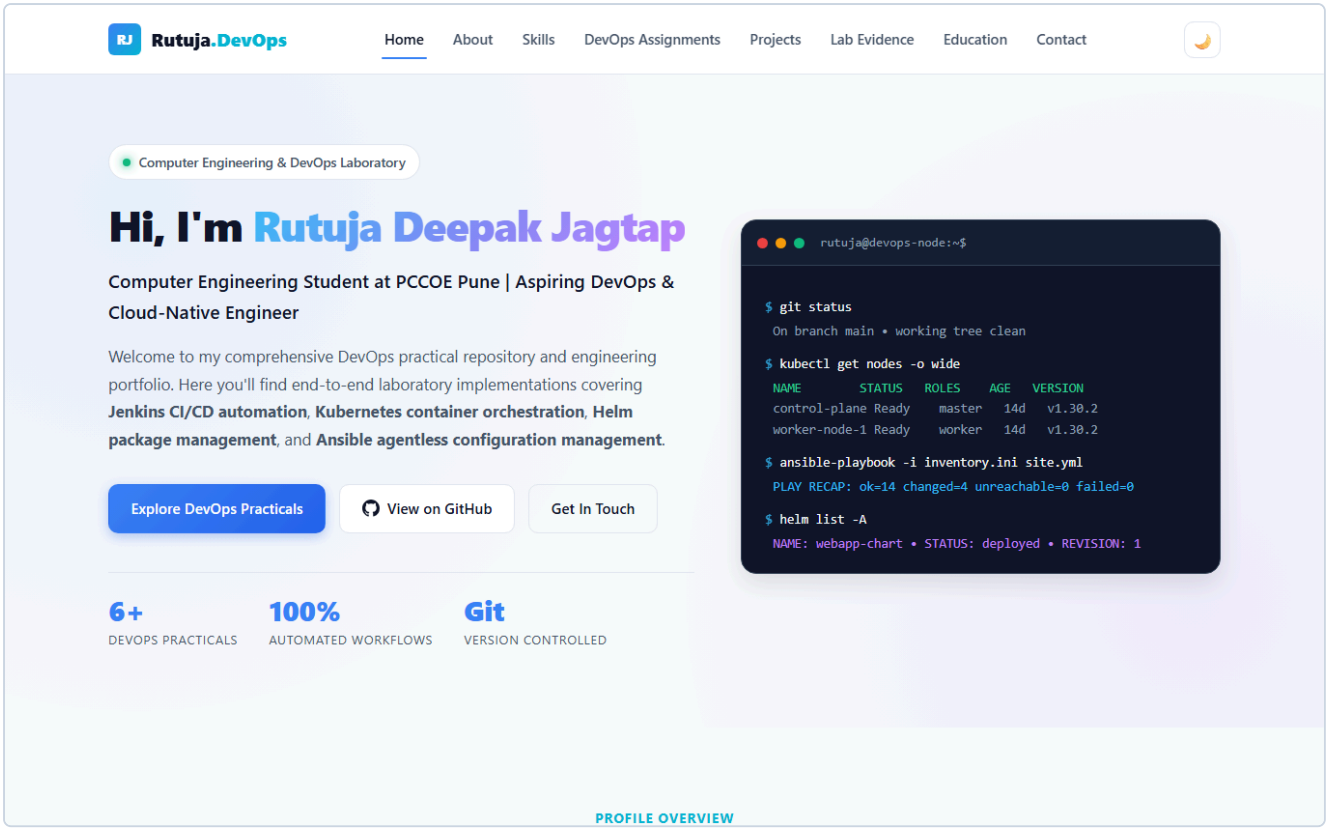


Figure 5: Desktop view of the portfolio homepage featuring animated terminal card, navigation, and student bio.

Proof 6: Live Personal Portfolio Website (Light Theme)

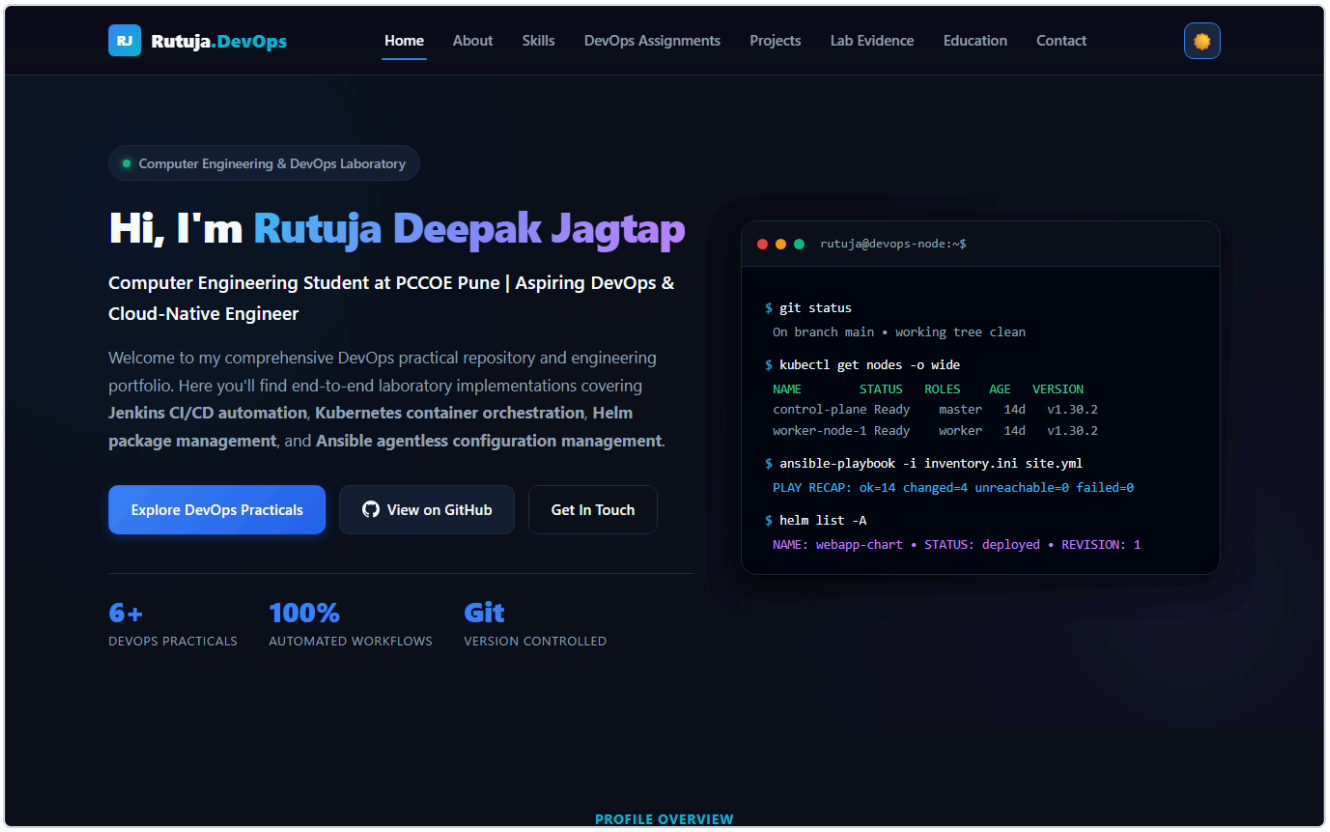


Figure 6: Demonstration of the responsive light theme toggle with high-contrast accessibility typography.

Proof 7: DevOps Practical Assignments Grid with Category Filters

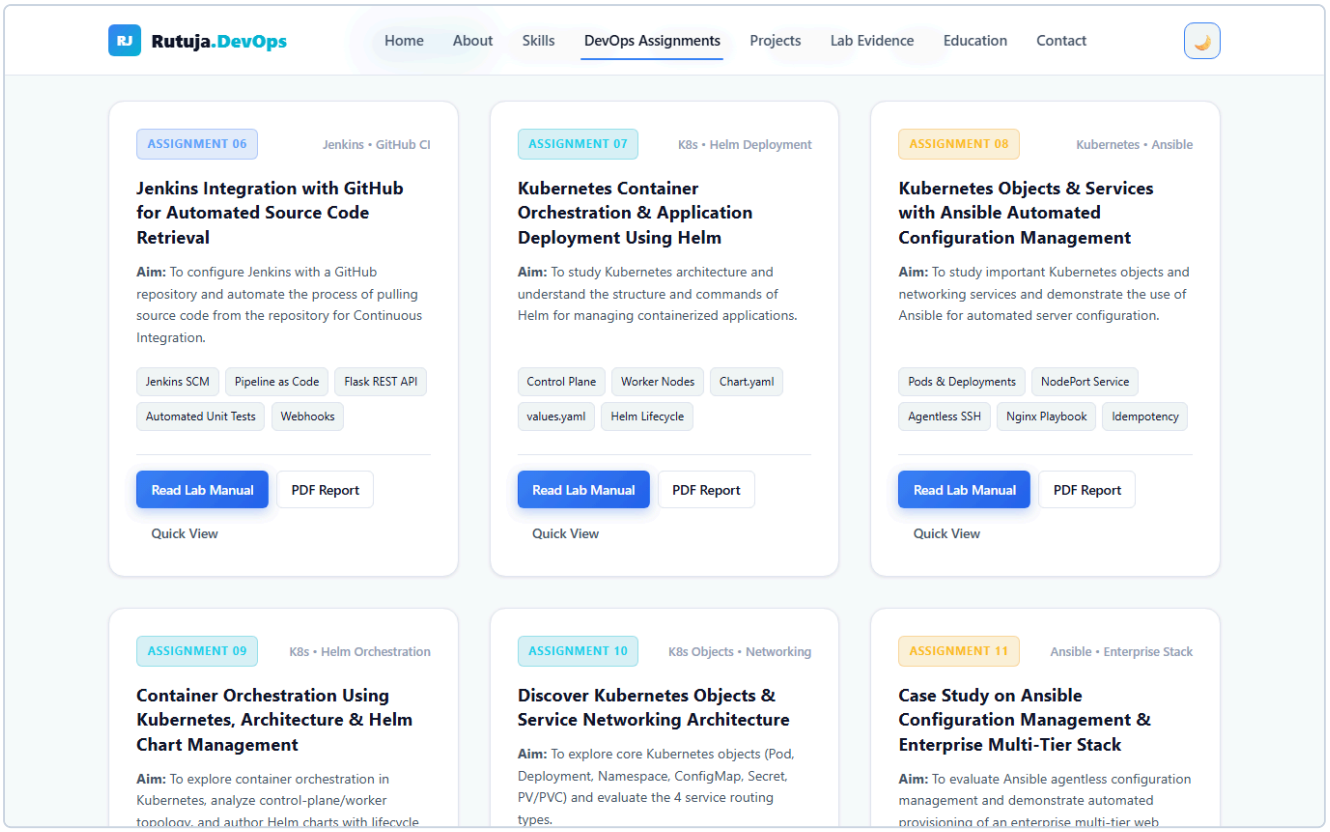


Figure 7: Filterable practical cards (All, CI/CD, Kubernetes, Helm, Ansible) with direct links to manuals and PDFs.

Proof 8: Interactive Quick View Modal with One-Click Command Copy

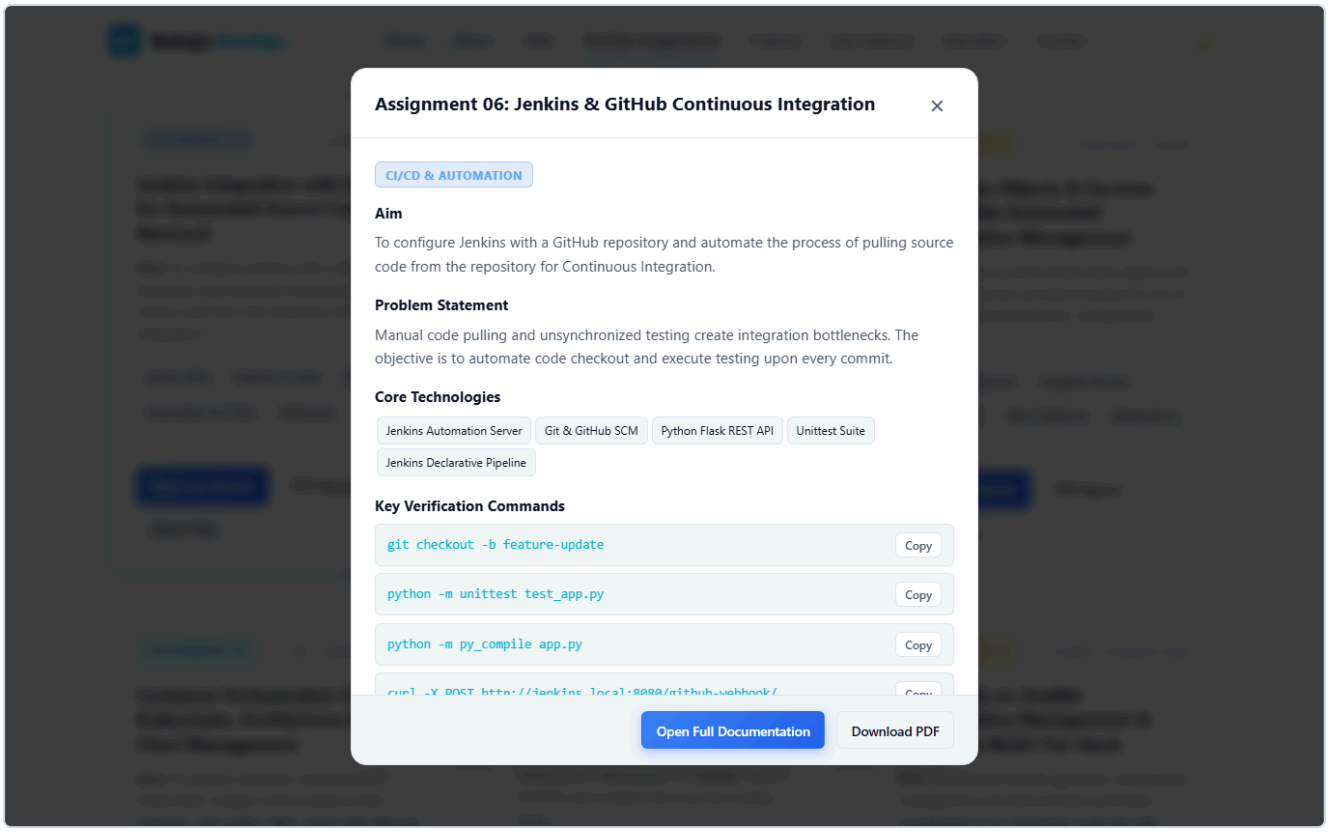


Figure 8: Modal dialog rendering assignment details, verification commands, and copy-to-clipboard buttons with toast alert.

Proof 9: Mobile Viewport Responsive Layout (iPhone 390x844)


RJ
Rutuja.DevOps
Computer Engineering & DevOps Laboratory
Hi, I'm Rutuja Deepak Jagtap
Computer Engineering Student at PCCOE Pune | Aspiring DevOps & Cloud-Native Engineer
Welcome to my comprehensive DevOps practical repository and engineering portfolio. Here you'll find end-to-end laboratory implementations covering Jenkins CI/CD automation, Kubernete
container orchestration, Helm package
management, and Ansible agentless

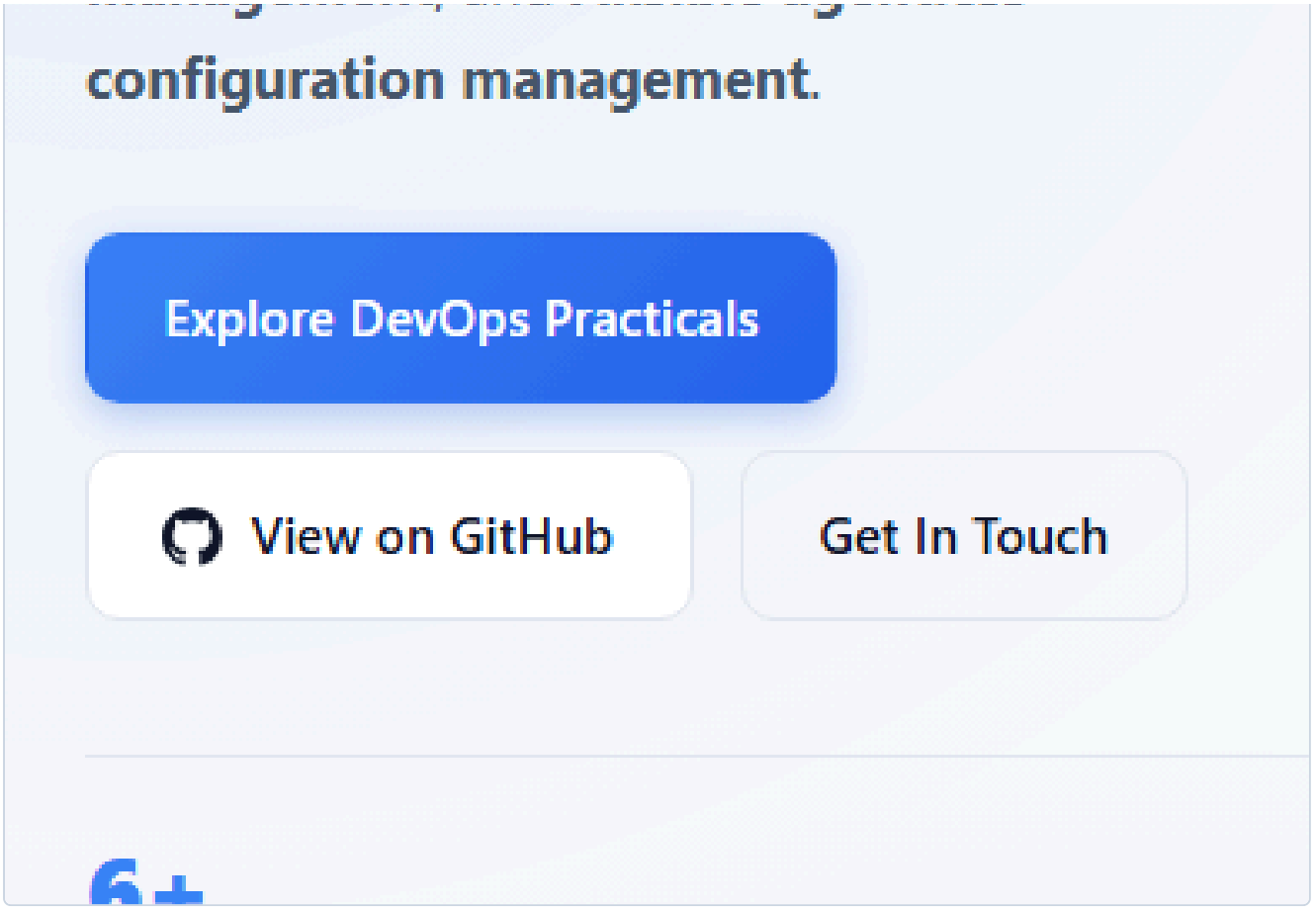


Figure 9: Fully responsive layout tested on mobile viewports with collapsible touch-friendly hamburger drawer.

7. Practical Verification Checklist

Checklist Item	Verification Result
Git repository initialized with clean .gitignore	**PASSED** (Zero build artifacts or temporary OS files)
Semantic atomic commit history	**PASSED** (chore, feat, docs, fix, merge)
Feature branch & non-fast-forward merge	**PASSED** (Branch `feature/portfolio-enhancements` merged)
Public GitHub remote synchronization	**PASSED** (Pushed to `rutujadeepakjagtap/DevOps`)
Assignments 06 to 11 organized with code	**PASSED** (Jenkinsfile, K8s YAMLs, Helm charts, playbooks)
Original 6 college PDF documents preserved	**PASSED** (Stored in `assets/docs/` and root)
Portfolio website responsiveness & themes	**PASSED** (Mobile, tablet, desktop, dark & light themes)
Interactive features & copy-to-clipboard	**PASSED** (Category filter, modal viewer, clipboard toast)
GitHub Pages static deployment ready	**PASSED** (Root index.html with relative asset references)

8. Result

The public DevOps practical repository was successfully engineered, version controlled using Git, and published to GitHub. The responsive portfolio website showcases student profile details, academic background, and practical

assignments with full runnable code manifests, architecture blueprints, and verification evidence.

9. Conclusion

Through this laboratory assignment, mastery of Git version control principles, feature branching, responsive web design without external frameworks, and static hosting via GitHub Pages was demonstrated. The repository stands as an authentic, production-grade engineering portfolio for academic evaluation at PCCOE Pune.